



Association of economic and racial segregation with restricted buprenorphine dispensing in U.S. community pharmacies

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HIGHLIGHTS

- Racial and economic inequities may persist at the point of dispensing.
- Joint racial and economic segregation was associated with restricted buprenorphine dispensing.
- Considering joint racial and economic segregation may uncover treatment gaps.
- Interventions to address pharmacy-level barriers are needed.

ARTICLE INFO

Keywords:

Opioid use disorder
Pharmacoequity
Structural racism

ABSTRACT

Background: Timely and reliable dispensing of buprenorphine is critical to accessing treatment for opioid use disorder (OUD). Racial and ethnic inequities in OUD treatment access are well described, but it remains unclear if inequities persist at the point of dispensing.

Methods: We analyzed data from a U.S. telephone audit that measured restricted buprenorphine dispensing in community pharmacies, defined as inability to fill a buprenorphine prescription requested by a “secret shopper.” Using the Index of Concentration at the Extremes (ICE), we constructed county-level measures of racial, ethnic, economic, and racialized economic (joint racial and economic segregation) segregation. Logistic regression models evaluated the association of ICE measures and restricted buprenorphine dispensing, adjusting for county type (urban vs. rural) and pharmacy type (chain vs. independent).

Results: Among 858 pharmacies surveyed in 473 counties, pharmacies in the most ethnically segregated and economically deprived counties had 2.66 times the odds (95 % CI: 1.41, 5.17) of restricting buprenorphine dispensing, compared to pharmacies in the most privileged counties after adjustment. Pharmacies in counties with high racialized economic segregation (quintile 2 and 3) also had higher odds of restricting buprenorphine dispensing (aOR 3.09 [95 % CI 1.7, 5.59]; aOR 2.11 [95 % CI 1.17, 3.98]). Similar associations were observed for economic segregation (aOR: 2.18 [95 % CI: 1.21, 3.99]), but not ethnic (0.59 [0.34, 1.05]) or racial (0.61 [0.35, 1.07]) segregation alone.

Conclusions: Restricted buprenorphine dispensing was most pronounced in socially and economically disadvantaged communities, potentially exacerbating gaps in OUD treatment access. Policy interventions should target both prescribing and dispensing capacity to advance pharmacoequity.

Abbreviations: ACS, American Community Survey; aOR, Adjusted odds ratio; CI, Confidence interval; DEA, Drug Enforcement Administration; ICE, Index of concentration at the extremes; MOTAA, Modernizing Opioid Treatment Access Act; MOUD, Medication for opioid use disorder; OR, Odds ratio; OTP, Opioid treatment program; OUD, Opioid use disorder.

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<https://doi.org/10.1016/j.dadr.2024.100255>

Received 21 February 2024; Received in revised form 8 May 2024; Accepted 28 June 2024

Available online 14 July 2024

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1. Introduction

The overdose crisis in the U.S. continues to accelerate, fueled by synthetic opioids and polysubstance use (Gomes et al., 2023). Racial and ethnic disparities have been observed, as overdose fatality rates nearly tripled for Black and Hispanic/Latinx Americans in the past decade, compared to a 58 % increase among non-Hispanic White individuals (Friedman et al., 2022, 2021; Romero et al., 2023). These grave racial and ethnic inequities stem from the forces of structural racism, which influence both policy and healthcare delivery, including access to medications for opioid use disorder (MOUD) (Jackson et al., 2022).

MOUD has been heavily racialized and is best understood as a competing frame between treating opioid use disorder (OUD) as a matter of crime control versus a biomedical condition. Methadone emerged in the 1960s as a response to urban crime, with its effectiveness defined in one of the earliest clinical studies by two social indicators: employment and arrests at six months (Dole et al., 1966; Hansen et al., 2023). Diversion of methadone was a primary concern from the onset, and thus the Drug Enforcement Administration (DEA) required clinics to incorporate strict surveillance and regulatory measures, including daily observed dosing and routine urine toxicology testing (Hansen et al., 2023). Methadone treatment began in an era marked by racial unrest and white flight from diversifying urban centers, fomenting white residents' fears that opioid treatment programs (OTPs) would bring an influx of Black and Latinx patients with OUD to predominantly white communities (Hansen et al., 2023; Hansen and Roberts, 2012). As a result, OTPs were frequently established in politically disempowered Black and Latinx neighborhoods (Hansen et al., 2023; Hansen and Roberts, 2012), and this racialized geographic distribution of methadone treatment capacity persists today (Goedel et al., 2020).

Buprenorphine, by contrast, emerged during the first wave of the overdose crisis that was driven by prescription opioids (Ciccarone, 2019), and successful uptake required distinguishing it from methadone by race and class (Hansen et al., 2023). This was accomplished by approving buprenorphine for office-based prescribing and language that implicitly identified white, middle-class patients, who were frequently covered by commercial insurance, as buprenorphine's targeted market (Hansen et al., 2023; Hansen and Netherland, 2016; Hatcher et al., 2018). As a result, access to buprenorphine remains starkly divided along social axes, with white patients having roughly four times the odds of receiving buprenorphine, compared to Black patients (Lagisetty et al., 2019). Among those with public insurance, minoritized racial and ethnic patients with OUD are significantly less likely to receive MOUD (Dunphy et al., 2022; Barnett et al., 2023).

The OUD cascade of care has been put forth as a public health framework to improve OUD treatment, spanning prevention, diagnosis, linkage to care, MOUD initiation, retention for ≥ 6 months, and recovery (Williams et al., 2022). The racialized MOUD landscape contributes to a staggering treatment gap, with as few as 13 % of individuals diagnosed with OUD receiving pharmacotherapy (Krawczyk et al., 2022). While there have been notable changes to improve MOUD access and ameliorate racial disparities in MOUD over the past decade, such as allowing advanced practice providers to prescribe buprenorphine (Comprehensive Addiction and Recovery Act, 2016) and elimination of the X-waiver (Consolidated Appropriation Act, 2023), interventions have focused almost exclusively on prescribing capacity (Lee et al., 2021; Fiscella et al., 2019; Incze and Garland, 2023), overlooking the importance of dispensing capacity (H. L. F. Cooper et al., 2020; Cooper et al., 2020).

Community pharmacies are increasingly implicated in dispensing practices that undermine buprenorphine availability (H. L. F. Cooper et al., 2020; Cooper et al., 2020; Weiner et al., 2023). For one, pharmacy deserts—defined as communities with low geographic access to pharmacy services—are disproportionately more common in Black and Hispanic/Latinx neighborhoods (Guadamuz et al., 2021). Additionally, three separate “secret shopper” studies have found that roughly

one-third of community pharmacies are unwilling to order and dispense buprenorphine, with estimates ranging from 27–38 % (Hill et al., 2022, 2021; Kazerouni et al., 2021). In addition to being unwilling to dispense, some pharmacies may not regularly stock buprenorphine, creating a delay of 1–4 days from when a prescription is written and the patient accesses the medication (Hill et al., 2022). Timely and reliable dispensing of buprenorphine is critical to ensuring sustained access to OUD treatment, as even short delays have been shown to impede retention (Lee et al., 2019).

Although dispensing restrictions may lead to inequities in MOUD, and a growing literature base has highlighted the role of social and economic inequities in behavioral health services (Barnett et al., 2023; Goedel et al., 2020; Hansen et al., 2023, 2013), past studies have not explored how social and economic privilege are associated with pharmacy-level barriers to MOUD. This study aims to address gaps in prior research by investigating the associations between county-level proxies of racism and classism with restricted buprenorphine dispensing in community pharmacies. Recognizing structural racism is maintained by the interaction between social and economic forces across institutions (Dean and Thorpe, 2022), we apply an intersectional lens to consider how the concurrent realities of racial segregation and economic deprivation may explain barriers to accessing buprenorphine from community pharmacies.

2. Materials and methods

2.1. Sample

To conduct this secondary analysis, we used data from a recent telephone audit (“secret shopper”) study that evaluated dispensing of prescribed buprenorphine (Kazerouni et al., 2021). Sampling methods have been previously described (Kazerouni et al., 2021). In short, stratified random sampling was conducted, whereby investigators identified U.S. counties with overdose rates exceeding the national average ($n=473$). In each of the counties identified, one chain pharmacy and one independent pharmacy were randomly sampled, where possible, from the National Council for Prescription Drug Programs DataQ Pharmacy database (DataQ, 2020). Trained study staff made scripted telephone calls to each pharmacy in June 2020 to mimic a patient inquiry about filling a prescription for buprenorphine. Pharmacy responses were categorized in one of three groups: (a) can dispense, (b) cannot dispense, or (c) unwilling to disclose buprenorphine availability over the telephone (Kazerouni et al., 2021).

Pharmacies that were unwilling to disclose buprenorphine availability were excluded from this analysis ($n=63$), resulting in an analytic sample of 858 pharmacies. We then geocoded pharmacy addresses using the tidygeocoder package in R to generate Federal Information Processing Standards (FIPS) codes for the state and county associated with each address (Cambon et al., 2021). This study was exempt by the Institutional Review Boards (IRB) at Oregon Health and Science University and Oregon State University. Reporting adheres to Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for observational studies.

2.2. Measures

Our primary outcome was restricted buprenorphine dispensing, defined as a pharmacy staff member's report that they could not fill the buprenorphine prescription requested by the secret shopper, where $Y=0$ indicates can dispense and $Y=1$ indicates cannot dispense.

Measures of segregation were operationalized using the Index of Concentration at the Extremes (ICE), which measures social spatial polarization. ICE confers advantages over other measures of social and economic inequality, in that ICE scores can distinguish segregated populations of extreme privilege (values closer to +1) from those of extreme deprivation (values closer to -1) (Krieger et al., 2016).

Additionally, ICE scores can be used to assess racial segregation and economic segregation independently or in combination (Feldman et al., 2015; Krieger et al., 2016).

For this study, we used ICE to construct the following exposures of interest: (1) racial segregation, (2) ethnic segregation, (3) economic segregation, and (4) racialized economic segregation (Feldman et al., 2015; Krieger et al., 2016; Massey, 2001). Two distinct measures of racialized economic segregation were constructed, one of which jointly considers race (Black vs. white) and income and the other jointly considers ethnicity (Hispanic or Latinx vs. non-Hispanic white) and income.

ICE scores were computed with the following formula: $ICE_i = (A_i - P_i)/T_i$, where i represents the geographic unit of analysis (i.e., county), A_i represents the number of persons in the most racially and/or economically privileged quintile in area i , P_i represents the number of persons in the most racially and/or economically disadvantaged quintile in area i , and T_i represents the total population in area i (Feldman et al., 2015; Krieger et al., 2016). A race ICE score was computed to assess racial segregation, where A_i represents the number of non-Hispanic white households, and P_i represents the number of non-Hispanic Black households (Feldman et al., 2015). Similarly, for ethnicity ICE, A_i represents the number of non-Hispanic white households and P_i represents the number of Hispanic or Latinx households (Wise et al., 2023). For income ICE, A_i represents the number of households with annual incomes above the 80th percentile, and P_i represents the number of households with an annual income below the 20th percentile (Feldman et al., 2015). Racialized economic segregation was assessed by computing (a) a joint race and income ICE, where A_i represents the number of non-Hispanic White households with annual incomes above the 80th percentile, and P_i represents the number of Black households with annual incomes below the 20th percentile (Feldman et al., 2015), and (b) a joint ethnicity and income ICE, where A_i represents the number of non-Hispanic White households with annual incomes above the 80th percentile, and P_i represents the number of Hispanic or Latinx households with annual incomes below the 20th percentile (Wise et al., 2023). Income percentile thresholds were \$125,000 and \$25,000, based on proximity to published national estimates of income quintiles (\$130,545 and \$26,685, respectively) in the 2020 American Community Survey (ACS) Table B19080 (U.S. Census Bureau, 2020; Wise et al., 2023). Data for race ICE and ethnicity ICE were drawn from 2020 ACS Table B03002 (U.S. Census Bureau, 2020). Data for the income ICE, joint race and income ICE, and joint ethnicity and income ICE were drawn from the 2020 ACS Table B19001 (U.S. Census Bureau, 2020). All ACS data used were 5-year estimates (2016–2020), for which the U.S. Census Bureau revised its weighting procedures to account for the impact of the COVID-19 pandemic (ACS and COVID-19., 2022).

All ICE scores were constructed at the county-level and were linked by 5-digit FIPS codes consisting of concatenated state and county codes. ICE scores were treated as categorical variables, using sample data to establish quintiles. Quintile 5 represents pharmacies in the most privileged counties, and quintile 1 represents the most deprived counties.

2.3. Statistical methods

We developed logistic regression models, with cluster-robust standard errors to account for potential clustering within counties, to evaluate the association between county-level social spatial polarization (measures of segregation) and restricted buprenorphine dispensing (cannot dispense vs. can dispense) in community pharmacies. For all models, quintile 5 (most privileged) served as the reference group.

We first developed one overall model, followed by two models stratified by independent and chain pharmacies, as buprenorphine dispensing restrictions are more common in independent pharmacies, compared to chain pharmacies (Hill et al., 2022, 2021; Kazerouni et al., 2021). Tests for interaction between ICE scores and pharmacy type were conducted.

Adjustments were made for potential confounders, including

geographic region (Midwest, Northeast, South, and West) and county type (urban vs. rural). In the overall model, adjustments were made for pharmacy type (chain vs. independent). All analyses were performed using R Statistical Software, version 4.3.1 (R Foundation for Statistical Computing).

3. Results

The distribution of pharmacies across ICE quintiles is shown in Table 1. Among the 858 pharmacies included in our study sample, 675 (78.7 %) dispensed buprenorphine and 183 (21.3 %) restricted buprenorphine dispensing. Pharmacies represented 41 states and 470 counties across the U.S. Sample characteristics are shown in Table 2. A plurality of sampled pharmacies were located in the South (41.4 %) and predominately urban counties (73.4 %). Pharmacies that were excluded due to non-response were comparable to pharmacies included in the analytic sample, in terms of distribution across ICE quintiles, geographic region, and county type, as shown in Table S1 in the supplementary material.

3.1. Racial segregation

County-level racial segregation (ICE race) was not associated with restricted buprenorphine dispensing (Table 3). The interaction between racial segregation and pharmacy type was not statistically significant ($P=0.671$), and no qualitative differences were observed between chain and independent pharmacies.

3.2. Ethnic segregation

A weak association was observed between county-level ethnic segregation (ICE ethnicity) and restricted buprenorphine dispensing (Table 3), where pharmacies located in counties with higher proportions of Hispanic or Latinx households had lower odds of restricted dispensing practices (quintile 4 odds ratio [OR]: 0.55 [95 % confidence interval (CI): 0.32, 0.95], $P=0.033$; quintile 2 OR: 0.60 [95 % CI: 0.37, 0.99],

Table 1
County-level ICE scores, by quintile.

	n	Range
ICE race		
Quintile 5	128	0.891, 0.969
Quintile 4	135	0.815, 0.891
Quintile 3	140	0.689, 0.815
Quintile 2	136	0.494, 0.689
Quintile 1	136	-0.613, 0.494
ICE ethnicity		
Quintile 5	134	0.888, 0.974
Quintile 4	136	0.810, 0.887
Quintile 3	131	0.695, 0.810
Quintile 2	133	0.532, 0.695
Quintile 1	141	-0.593, 0.532
ICE income		
Quintile 5	146	0.085, 0.419
Quintile 4	141	-0.006, 0.085
Quintile 3	138	-0.071, -0.006
Quintile 2	132	-0.129, -0.071
Quintile 1	118	-0.401, -0.129
ICE race x income		
Quintile 5	153	0.181, 0.430
Quintile 4	139	0.135, 0.181
Quintile 3	131	0.102, 0.135
Quintile 2	117	0.064, 0.102
Quintile 1	135	-0.243, 0.064
ICE ethnicity x income		
Quintile 5	131	0.183, 0.430
Quintile 4	136	0.143, 0.183
Quintile 3	138	0.110, 0.143
Quintile 2	132	0.085, 0.110
Quintile 1	138	-0.242, 0.085

Table 2
Characteristics of pharmacies sampled, stratified by dispensing outcome. ^a.

	Dispenses Buprenorphine (n=675)	Restricted Dispensing of Buprenorphine (n=183)	Overall (n=858)
County Type			
Urban	500 (79.4)	130 (20.6)	630 (100.0)
Rural	175 (76.8)	53 (23.2)	228 (100.0)
Race ICE			
Quintile 5	128 (74.0)	45 (26.0)	173 (100.0)
Quintile 4	135 (79.4)	35 (20.6)	170 (100.0)
Quintile 3	140 (81.9)	31 (18.1)	171 (100.0)
Quintile 2	136 (78.6)	37 (21.4)	173 (100.0)
Quintile 1	136 (79.5)	35 (20.5)	171 (100.0)
Ethnicity ICE			
Quintile 5	134 (77.9)	38 (22.1)	172 (100.0)
Quintile 4	136 (79.1)	36 (20.9)	172 (100.0)
Quintile 3	131 (77.1)	39 (22.9)	170 (100.0)
Quintile 2	133 (77.8)	38 (22.2)	171 (100.0)
Quintile 1	141 (81.5)	32 (18.5)	173 (100.0)
Income ICE			
Quintile 5	146 (85.4)	25 (14.6)	171 (100.0)
Quintile 4	141 (81.5)	32 (18.5)	173 (100.0)
Quintile 3	138 (80.7)	33 (19.3)	171 (100.0)
Quintile 2	132 (77.2)	39 (22.8)	171 (100.0)
Quintile 1	118 (68.6)	54 (31.4)	172 (100.0)
Race x Income ICE			
Quintile 5	153 (88.4)	20 (11.6)	173 (100.0)
Quintile 4	139 (81.8)	31 (18.2)	170 (100.0)
Quintile 3	131 (76.6)	40 (23.4)	171 (100.0)
Quintile 2	117 (68.0)	55 (32.0)	172 (100.0)
Quintile 1	135 (78.5)	37 (21.5)	172 (100.0)
Ethnicity x Income ICE			
Quintile 5	131 (76.6)	40 (23.4)	171 (100.0)
Quintile 4	136 (79.5)	35 (20.5)	171 (100.0)
Quintile 3	138 (80.2)	34 (19.8)	172 (100.0)
Quintile 2	132 (76.7)	40 (23.3)	172 (100.0)
Quintile 1	138 (80.2)	34 (19.8)	172 (100.0)
Geographic Region			
Midwest	192 (80.3)	47 (19.7)	239 (100.0)
Northeast	180 (85.7)	30 (14.3)	210 (100.0)
South	255 (71.8)	100 (28.2)	355 (100.0)
West	48 (88.9)	6 (11.1)	54 (100.0)
Counties	426	162	470
Represented States	41	27	41
Represented			

^aData are reported as n (row %).

P=0.044). Associations at these two quintiles persisted after adjusting for county type (urban vs. rural) and geographic region (quintile 4 adjusted odds ratio [aOR]: 0.52 [95 % CI: 0.30, 0.92], P=0.025; quintile 2 aOR: 0.56 [95 % CI: 0.32, 0.97], P=0.037). The interaction between ethnic segregation and pharmacy type was not statistically significant (P=0.982), with overlapping confidence intervals for analyses stratified by pharmacy type.

3.3. Economic segregation

County-level economic segregation (ICE income) was associated with restricted buprenorphine dispensing (Table 3). Pharmacies located in the most economically disadvantaged counties (i.e., quintile 1) had significantly higher odds (OR: 2.67 [95 % CI: 1.56, 4.58], P<0.001) of restricting buprenorphine dispensing, compared to counties characterized by greatest economic privilege (i.e., quintile 5). The significance of this association persisted (aOR: 2.18 [95 % CI: 1.17, 4.01], P=0.014) after adjustment. The interaction between economic segregation and pharmacy type was not statistically significant (P=0.718), with comparable magnitudes of association between chain and independent pharmacies.

3.4. Racialized economic segregation

Racialized economic segregation (ICE race x income), was associated with dispensing restrictions at the lowest three quintiles (i.e., quintiles 1–3) overall; however after adjusting for confounders, statistical significance persisted at quintile 3 (OR: 2.34 [1.29, 4.22], P=0.005; aOR: 2.11 [1.17, 3.98], P=0.014) and quintile 2 (OR: 3.60 [2.05, 6.30], P<0.001; aOR: 3.09 [1.70, 5.59], P<0.001), but not quintile 1. The interaction between racialized economic segregation and pharmacy type was not statistically significant (P=0.512), with similar associations observed among chain and independent pharmacies.

Similarly, when considering Hispanic or Latinx vs. non-Hispanic white differences (ICE ethnicity x income), racialized economic segregation was associated with dispensing restrictions at the four lowest quintiles, even after adjusting for confounders. The strongest association was observed at quintile 1 (OR: 3.10 [1.72, 5.57], P<0.001; aOR: 2.66 [1.38, 5.16], P=0.004). Again, the interaction between racialized economic segregation and pharmacy type was not statistically significant (P=0.300), with comparable patterns in stratified analyses.

4. Discussion

Our study findings cast light on another dimension of racial, ethnic, and economic inequities across the OUD cascade of care. This suggests that barriers persist beyond prescribing, as buprenorphine dispensing restrictions are associated with some measures of social spatial polarization, potentially compounding barriers in accessing treatment. While null and weak associations were observed with racial and ethnic segregation, respectively, racialized economic segregation was consistently the strongest predictor of buprenorphine dispensing barriers. These findings underscore the importance of considering an intersectional approach when investigating dimensions of MOUD access, recognizing the simultaneous and interlocking forms of structural oppression and discrimination that coincide to produce health inequities (Agénor, 2020; Feldman et al., 2015).

Significant effects of racialized economic segregation, after adjustment, were observed in quintiles 2 and 3 when considering Black vs. white differences and all four quintiles when considering Hispanic or Latinx vs. non-Hispanic white differences, whereas the association with economic segregation alone was only significant in the lowest quintile. It is unclear why the association with racialized economic segregation attenuates at the lowest (i.e., most deprived) quintile for Black vs. white differences, but not for Latinx vs. white differences. This may be the result of measurement error. While it is common to categorize ICE scores into quintiles (Chambers et al., 2019; Krieger et al., 2018, 2016), the cutoffs may establish arbitrary groups that are not homogenous, which may impact interpretation. Alternatively, dispensing restrictions may be least common in the most racially segregated communities due to the ethnic enclave effect, which refers to the social networks in these communities that are characterized by strong ties, social cohesion, and emotional support (Smith, 2024). Pharmacists are woven into this social fabric, with previous qualitative studies describing the strong ties between residents and pharmacists that practice in the communities where they grew up and understand the community's hardships (Cooper et al., 2020). Further research is needed to better explain the patterns observed in the present study.

Increased attention is now being paid to the impact of 'pharmacy deserts' (Guadamuz et al., 2021; Qato et al., 2014; Wisse et al., 2021; Ying et al., 2022). A recent investigation into the 30 most populous cities in the U.S. found that the density of pharmacies is consistently lower among highly segregated communities of color, compared to both diverse neighborhoods (i.e., those where no racial/ethnic group comprises ≥ 50 % of the population) and highly segregated white communities (Guadamuz et al., 2021). Even when patients can physically access pharmacies, these pharmacies generally have shorter business hours and are less likely to offer enhanced services (e.g., home delivery), further

Table 3

Association between county-level social spatial polarization and restricted buprenorphine dispensing in U.S. community pharmacies.

	Chain Pharmacy (N=418)		Independent Pharmacy (N=440)		Overall (N=858)	
	Unadjusted OR [95 % CI]	Adjusted OR [95 % CI]	Unadjusted OR [95 % CI]	Adjusted OR [95 % CI]	Unadjusted OR [95 % CI]	Adjusted OR [95 % CI]
Race ICE^a						
Quintile 5	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Quintile 4	0.60 [0.26 – 1.35]	0.56 [0.23 – 1.29]	0.81 [0.42 – 1.55]	0.87 [0.44 – 1.70]	0.74 [0.44 – 1.25]	0.75 [0.43 – 1.29]
Quintile 3	0.61 [0.27 – 1.35]	0.58 [0.24 – 1.37]	0.64 [0.32 – 1.26]	0.70 [0.33 – 1.47]	0.63 [0.37 – 1.08]	0.66 [0.37 – 1.17]
Quintile 2	0.74 [0.34 – 1.61]	0.66 [0.28 – 1.56]	0.79 [0.41 – 1.52]	0.85 [0.41 – 1.76]	0.77 [0.46 – 1.29]	0.78 [0.43 – 1.40]
Quintile 1	0.67 [0.30 – 1.47]	0.49 [0.20 – 1.16]	0.78 [0.40 – 1.51]	0.71 [0.34 – 1.48]	0.73 [0.44 – 1.22]	0.61 [0.34 – 1.12]
Ethnicity ICE						
Quintile 5	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Quintile 4	0.63 [0.28, 1.38]	0.55 [0.24, 1.23]	0.49* [0.25, 0.96]	0.49* [0.24, 0.99]	0.55* [0.32, 0.95]	0.52* [0.30, 0.92]
Quintile 3	0.46 [0.19, 1.05]	0.41* [0.16, 0.98]	0.84 [0.44, 1.58]	0.90 [0.45, 1.78]	0.67 [0.40, 1.12]	0.67 [0.40, 1.15]
Quintile 2	0.72 [0.33, 1.53]	0.61 [0.27, 1.39]	0.52 [0.26, 1.02]	0.51 [0.24, 1.05]	0.60* [0.37, 0.99]	0.56* [0.32, 0.97]
Quintile 1	0.50 [0.22, 1.11]	0.43 [0.17, 1.05]	0.70 [0.36, 1.33]	0.73 [0.34, 1.52]	0.61 [0.37, 1.03]	0.59 [0.32, 1.09]
Income ICE						
Quintile 5	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Quintile 4	0.72 [0.30 – 1.70]	0.75 [0.31 – 1.80]	2.23* [1.02 – 5.13]	1.90 [0.85 – 4.43]	1.33 [0.74 – 2.38]	1.23 [0.69 – 2.23]
Quintile 3	0.44 [0.16 – 1.13]	0.49 [0.17 – 1.30]	3.04* [1.42 – 6.89]	2.48* [1.12 – 5.80]	1.40 [0.80 – 2.44]	1.29 [0.72 – 2.34]
Quintile 2	0.97 [0.42 – 2.22]	1.13 [0.46 – 2.76]	2.76* [1.30 – 6.23]	2.16 [0.96 – 5.07]	1.73 [0.96 – 3.09]	1.57 [0.85 – 2.85]
Quintile 1	1.98 [0.95 – 4.28]	2.15 [0.90 – 5.29]	3.57** [1.70 – 7.97]	2.47* [1.09 – 5.89]	2.67*** [1.56 – 4.58]	2.18* [1.17 – 4.01]
Race x Income ICE						
Quintile 5	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Quintile 4	1.47 [0.59 – 3.79]	1.45 [0.55 – 3.88]	1.90 [0.86 – 4.40]	1.55 [0.68 – 3.65]	1.71 [0.91 – 3.19]	1.51 [0.81 – 2.82]
Quintile 3	1.37 [0.55 – 3.53]	1.39 [0.53 – 3.74]	3.45** [1.62 – 7.79]	2.83* [1.27 – 6.60]	2.34** [1.29 – 4.22]	2.11* [1.17 – 3.98]
Quintile 2	3.45** [1.53 – 8.40]	4.05** [1.62 – 10.84]	3.68*** [1.76 – 8.21]	2.67* [1.20 – 6.25]	3.60*** [2.05 – 6.30]	3.09*** [1.70 – 5.59]
Quintile 1	1.45 [0.58 – 3.74]	1.16 [0.42 – 3.34]	2.70* [1.36 – 6.11]	1.68 [0.73 – 4.05]	2.10* [1.15 – 3.83]	1.45 [0.74 – 2.82]
Ethnicity x Income ICE						
Quintile 5	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
Quintile 4	1.93 [0.82, 4.82]	1.96 [0.81, 5.00]	2.82* [1.28, 6.62]	2.38* [1.06, 5.67]	2.36** [1.30, 4.30]	2.18* [1.20, 3.97]
Quintile 3	1.24 [0.49, 3.24]	1.27 [0.48, 3.45]	3.67** [1.70, 8.50]	3.01* [1.33, 7.21]	2.35** [1.28, 4.30]	2.13* [1.14, 4.00]
Quintile 2	1.82 [0.75, 4.63]	2.02 [0.79, 5.38]	2.64* [1.21, 6.16]	2.04 [0.90, 4.91]	2.27** [1.24, 4.15]	2.01* [1.08, 3.75]
Quintile 1	2.38 [1.02, 5.88]	2.65 [1.01, 7.28]	3.80*** [1.77, 8.75]	2.76* [1.19, 6.79]	3.10*** [1.72, 5.57]	2.66** [1.38, 5.16]

^a Quintile 5 represents the most privileged and quintile 1 represents the most deprived.

* denotes P<0.05

** denotes P<0.01, and

*** denotes P<0.001

perpetuating inequities (Chisholm-Burns et al., 2017; Guadamuz et al., 2021). Pharmacy deserts are an enduring testament to the structurally racist processes of disinvestment and residential segregation, creating resource scarcity in Black and Latinx neighborhoods (Satcher, 2022).

It remains unclear why a notable proportion of pharmacies across the U.S. will not dispense buprenorphine, but the issue is likely multifactorial in nature (Qato et al., 2022), with barriers and facilitators to prompt dispensing at both the individual- and pharmacy-level (Hill et al., 2023). Pharmacists' attitudes may hinder access (Cooper et al., 2020; Thornton et al., 2017; Ventricelli et al., 2020), as stigma towards

patients taking buprenorphine is a key predictor of pharmacists' willingness to dispense (Light et al., 2024). However, there are other factors at play, including pharmacy-level policies, such as only dispensing buprenorphine when the prescriber is local (Hill et al., 2023), underscoring the importance of trust and communication between prescribers and pharmacists (Cooper et al., 2020; Major et al., 2023). Additionally, in a prior qualitative study in the Appalachian Midwest, pharmacists expressed concerns about potential investigations from the DEA for exceeding a "cap" on buprenorphine prescriptions (Cooper et al., 2020). While the DEA does not explicitly set distribution thresholds, or caps

(Ostrach et al., 2021), wholesalers are required to monitor dispensing of opioids, including buprenorphine (H. L. F. Cooper et al., 2020; Ostrach et al., 2021). Wholesalers have devised proprietary algorithms to monitor dispensing, and pharmacies that are flagged by the algorithm can have their medication supply frozen by the wholesaler (Cooper et al., 2020). Pharmacists' concerns about wholesaler monitoring are widespread and may partially explain restricted dispensing (Cooper et al., 2020; Ostrach et al., 2021), considering some pharmacies, particularly independently-owned pharmacies, have opted to not stock buprenorphine altogether in response to such monitoring (Hansen et al., 2023).

In the wake of the removal of the X-waiver requirement in 2023, DEA and HHS have sought to clarify with distributors that any imposed quantitative thresholds should not interfere with the increasing number of patients who are likely to require access to buprenorphine (Milgram et al., 2023). However, it is unclear if wholesalers have adjusted their systems for identifying suspicious orders and communicated these adjustments to pharmacies. Together, these findings emphasize that physical access to pharmacies cannot be conflated with reliable and prompt access to buprenorphine for OUD treatment.

Further work is needed to understand restricted buprenorphine dispensing, particularly in differentiating between dispensing barriers that stem from pharmacist-level perceptions and pharmacy-level policies, to develop effective solutions. Future studies should use individual-level data, such as pharmacy claims, to evaluate the relationship between measures of social spatial polarization and more granular measures of buprenorphine availability. Additionally, qualitative studies are needed to better understand why community pharmacies located in racially or ethnically segregated and economically deprived counties restrict dispensing to help parse out the multiple contributing factors. Such research should be coupled with (a) educational interventions to reduce stigma associated with MOUD among pharmacists (Green et al., 2022; Irwin et al., 2023) and (b) monitoring of pharmacies that limit or refuse to fill buprenorphine prescriptions to inform policy interventions that target dispensing capacity. Finally, if access remains challenging, state-level legislation that mandates pharmacies both carry and dispense buprenorphine may be warranted (Qato et al., 2022).

This study has several limitations. First, this was a cross-sectional study conducted in June 2020, using 2018 data to identify high overdose burden counties. Geographic changes in overdose have been observed in recent years with the rise in polysubstance use (Friedman and Shover, 2023; Saunders et al., 2023) and stressors induced by the COVID-19 pandemic (Friedman and Akre, 2021). Our study sample, therefore, may not include some counties with high overdose burdens in recent years, and it is not a nationally representative sample of community pharmacies. Second, three-fourths of the pharmacies sampled were located in urban counties, and this lack of rural representation may bias measures of association. Third, there are measurement limitations associated with our exposure of interest. All ICE scores were constructed with household data from the ACS, which excludes persons not residing in housing units (Wise et al., 2023), and thus, may bias measures of county-level privilege and deprivation. Finally, our measure of restricted dispensing does not capture delays due to stocking challenges or depleted inventory, which could be a barrier to timely MOUD access. Thus, associations observed herein reflect only one dimension to buprenorphine availability and likely underestimate the effects of racialized economic segregation, as counties without access to a pharmacy were not included.

Despite its limitations, this study has critical implications for ongoing policy advances, such as the growing acceptance of telehealth to reach patients in rural areas with fewer buprenorphine prescribers (Frost et al., 2022), that may be blunted by restricted buprenorphine dispensing (Hill et al., 2023; Pednekar and Peterson, 2018). This is particularly relevant in the context of ongoing discussions about methadone reform vis-à-vis the Modernizing Opioid Treatment Access Act (MOTAA) (Markey,

2023), which would allow office-based addiction specialty providers to prescribe methadone that can be dispensed in community pharmacies, instead of restricting methadone dispensing to OTPs. This would allow methadone treatment for OUD within community-based practice, as is already done in many other countries (Calcaterra et al., 2019). Emerging research has demonstrated how such regulatory changes could bolster methadone prescribing capacity (Joudrey et al., 2023), as well as reduce drive times to treatment (Joudrey et al., 2020). While such reform is critically important, findings from our study warrant caution towards policy implementations that address prescribing alone, as barriers to MOUD dispensing, and thus inequities in OUD treatment access, will persist if specific provisions addressing pharmacy access barriers are not also incorporated into law.

5. Conclusions

To address inequities that permeate across the OUD cascade of care, it is critical to advance pharmacoequity, defined as “access to the highest-quality medications for all individuals, regardless of race, ethnicity, socioeconomic status, or availability of resources” (Chalasan et al., 2022; Essien et al., 2021). This orientation towards pharmacoequity amplifies the recommendations raised by other scholars to ensure that changes in policy and practice to increase prescribing of MOUD are accompanied by efforts to increase dispensing (Cooper et al., 2020), which requires greater recognition of the role of community pharmacies within the healthcare ecosystem.

Unequal access to OUD treatment is pervasive across the cascade of care, from prescribing to dispensing for those from racial/ethnic minoritized backgrounds and economically disadvantaged groups. Findings from this study demonstrate that proximity to a community pharmacy, while necessary, is likely not sufficient to ensuring MOUD treatment availability, as evidenced by the association between racialized economic segregation and restricted buprenorphine dispensing. Barriers to buprenorphine dispensing are most pronounced in socially and economically disadvantaged communities, further exacerbating the treatment access gaps these communities already experience. These findings have critical policy implications, in that interventions must aim to strengthen capacity in both prescribing and dispensing to achieve equitable access to treatment. Against the backdrop of a worsening overdose crisis, multilevel interventions are sorely needed to address persistent barriers to MOUD and advance pharmacoequity.

Funding

KJM was supported by an internal training grant (Health Equity Scholars Program) at the Johns Hopkins Bloomberg School of Public Health. XAL was supported by the Agency for Healthcare Research and Quality (K12 HS026370). Neither funder had any role in study design, writing, preparation of the article, or the decision to submit the article for publication. The content is solely the responsibility of the authors and does not necessarily represent the official views of the Agency for Healthcare Research and Quality.

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Ximena A. Levander: Writing – review & editing, Supervision, Data curation. **Sabriya L. Linton:** Writing – review & editing, Supervision, Methodology. **Neda J. Kazerouni:** Writing – review & editing, Investigation, Data curation. **Kyle J. Moon:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Adriane N. Irwin:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Daniel M. Hartung:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Kyle Moon reports financial support was provided by Johns Hopkins University Bloomberg School of Public Health's Health Equity Scholars Program. Ximena Levander reports financial support was provided by Agency for Healthcare Research and Quality (K12 HS026370). Adriane Irwin reports a relationship with Rx Drug Abuse & Heroin Summit that includes: speaking and lecture fees. Adriane Irwin reports a relationship with American Pharmacists Association that includes: travel reimbursement. Daniel Hartung reports a relationship with Alkermes Inc that includes: consulting or advisory. Serving as an unpaid member of the American College of Physicians health policy committee – XAL, Serving as unpaid grants and research committee chair for the National Coalition to Liberate Methadone – XAL, Member of the Oregon Health Authority commission (HERC) that oversees the medical benefit within the state's Medicaid program – ANI. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

None.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.dadr.2024.100255](https://doi.org/10.1016/j.dadr.2024.100255).

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